Weather-Based Crop Advisory System

Requirement Analysis Report

Academic Project Documentation

1. Introduction

Agriculture is strongly affected by weather conditions such as rainfall, temperature, humidity and wind. Farmers may receive weather forecasts, but a general forecast does not directly explain what action should be taken for a particular crop. The Weather-Based Crop Advisory System is proposed as a simple decision-support application that connects weather information with crop information and produces practical advisory messages.

2. Problem Statement

Farm activities such as irrigation and spraying can be affected by short-term weather changes. A farmer may need to interpret several weather values before deciding whether to continue, postpone or change an activity. The proposed system addresses this gap by using selected location, crop and crop-stage information together with weather conditions to generate understandable recommendations.

3. Existing System

- Weather information and agricultural information are often checked separately.
- General weather applications normally do not provide crop-specific actions.
- Manual interpretation of temperature, rain, humidity and wind can take time.
- Important forecast changes can be missed when the user does not check regularly.

4. Proposed System

The proposed system allows a farmer to log in, select a farming location and crop, view weather information and receive crop-specific advisories. An advisory engine evaluates predefined rules and displays recommendations related to irrigation, spraying, rainfall, temperature and other configured weather conditions. An administrator can maintain supported crops and advisory rules.

5. Objectives

- Provide location-based current and forecast weather information.
- Allow selection of crop and, where required, crop stage.
- Generate simple recommendations from weather conditions.
- Display useful alerts for weather situations affecting farm activities.
- Keep the interface simple and understandable.
- Keep advisory rules separate from the user interface.

6. Scope

The system covers user authentication, location selection, crop selection, weather retrieval, rule-based advisory generation and advisory display. Administrative functions cover crop and rule management. The system is a decision-support tool; it does not guarantee crop yield, replace an agricultural expert, diagnose diseases from images, or directly control farm equipment.

7. Stakeholders

Stakeholder	Role / Interest
Farmer	Uses weather information and advisories for farming decisions.
Administrator	Maintains crop data and advisory rules.
Developer	Builds, tests and maintains the application.
Weather Data Provider	Supplies weather observations and forecasts.
Agricultural Expert (optional)	Can review the agricultural logic used in advisory rules.

8. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in.
FR-02	The system shall allow a farmer to select or enter a farming location.
FR-03	The system shall allow the farmer to select a supported crop.
FR-04	The system shall accept crop-stage information where applicable.
FR-05	The system shall retrieve weather information for the selected location.
FR-06	The system shall evaluate weather data against active advisory rules.
FR-07	The system shall generate crop-specific recommendations.
FR-08	The system shall display relevant weather alerts.
FR-09	The administrator shall be able to manage crop information.
FR-10	The administrator shall be able to manage advisory rules.

9. Non-Functional Requirements

Category	Requirement
Usability	The interface should have clear labels, simple navigation and readable advice.
Performance	Normal operations should respond within a reasonable time for expected project usage.
Security	Authentication and role-based authorization should protect user and admin functions.
Reliability	Unavailable weather data should be handled without generating unsupported advice.
Maintainability	Weather integration and advisory logic should be modular.
Scalability	New crops and rules should be addable without redesigning the whole system.
Compatibility	A web implementation should support current mainstream browsers.

10. User Requirements

• A farmer should reach the main advisory with minimum navigation.

• Weather values should be shown with familiar units and labels.

• Advisories should explain the recommended action briefly.

• The system should clearly indicate when weather data is unavailable.

• Administrative operations should remain separate from farmer operations.

11. System Requirements

Type	Suggested Requirement
Client	Desktop, laptop, tablet or smartphone.
Browser	Current Chrome, Edge, Firefox or another modern browser.
Server	Standard development/server environment suitable for the chosen backend.
Database	Relational or document database depending on implementation.
Internet	Required for live external weather data.
Development Tools	IDE/editor, Git and selected runtime/framework.

12. Feasibility Analysis

Technical: The system can be developed with commonly available web technologies and an external weather API. Rule-based advisory logic is manageable for a student project.

Economic: The prototype can use free/open-source development tools and a weather service plan appropriate for development usage.

Operational: The workflow of selecting location and crop and reading the resulting advice is straightforward for the intended user.

13. Assumptions

• The weather service provides reasonably accurate and available data.

• Users enter correct crop and location information.

• Configured advisory rules are reviewed before use.

• The first version supports a limited set of crops and rules.

14. Constraints

• External weather API availability and usage limits may affect the application.

• Forecasts are predictions and can change.

• Advisory rules may need regional and crop-specific refinement.

15. Acceptance Criteria

1. A registered user can log in successfully.

2. The farmer can select a location and crop.

3. Weather information is displayed when the weather service is available.

4. A configured weather condition produces a relevant advisory.

5. The system gives a clear error when weather data is unavailable.

6. Unauthorized users cannot access administrator-only functions.

16. Summary

The requirements define a focused weather-to-advisory decision-support system. The scope is intentionally manageable for an academic project while allowing future additions such as more crops, notifications, regional rules and additional agricultural data.